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John B. McBride; Edward Ott; Jay P. Boris; Joseph H. Orens



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Theory and Simulation of Turbulent Heating by the Modified Two-Stream Instability

JOHN B. MCBRIDE, EDWARD OTT,* JAY P. BORIS, AND JOSEPH H. ORENS

Naval Research Laboratory, Washington, D. C. 20390

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Results of an analytical and numerical study of the nonresonant, modified plasma two-stream instability, which is driven by relative streaming of electrons and ions across a magnetic field $\mathbf{B}_0$ are presented. The instability has characteristic frequency and growth rate comparable to the lower-hybrid frequency. The linear theory is discussed both in the electrostatic and fully electromagnetic cases, and a detailed numerical study of the dependence of the unstable roots of the dispersion relation for a wide range of plasma parameters is presented. The nonlinear theory includes discussions of (1) quasilinear theory, (2) trapping, which is responsible for nonlinear stabilization, (3) a derivation of a fully nonlinear scaling law which shows how results scale with electron-ion mass ratio, and (4) the effect of cross-field vortex-like motion caused by turbulence induced $\mathbf{E} \times \mathbf{B}$ drifts. One- and two-dimensional computer simulations with dense k -space spectra are presented in support of this theory. The simulations show that the instability can be a very important turbulent heating mechanism that heats the ions (perpendicular to $\mathbf{B}_0$) and the electrons (parallel to $\mathbf{B}_0$) comparably. The final state has $(T_{i\perp}/m_i)^{1/2} \approx (T_{e\parallel}/m_e)^{1/2} \sim \frac{1}{2}U$, where U is the initial relative drift speed. Applications to experimental situations are discussed.

I. INTRODUCTION

Considerable effort has been devoted to the understanding of plasma instabilities driven by relative streaming between electrons and ions perpendicular to a magnetic field. Interest in such instabilities derives from the crucial role they play in anomalous resistance and plasma heating in situations of practical importance. As an example which has been treated in great detail because of its shock wave applications, we mention the beam-cyclotron instability,¹⁻³ a micro-instability of the electron cyclotron modes.

In this paper we present a detailed theory and extensive computer simulations of another instability driven by currents across a magnetic field $\mathbf{B}_0$ known as the "modified two-stream instability" (the instability has also been referred to as the lower-hybrid two-stream instability).⁴⁻¹² This name derives from the fact that, in the fluid limit, the form of the dispersion relation is analogous to that for the usual Buneman¹³ two-stream instability. As we shall see in Sec. II, however, there are important differences introduced by the magnetic field. First, the instability threshold is $U \gtrsim v_i$ rather than $U > v_e$, where U , $v_i = (T_i/m_i)^{1/2}$ and $v_e = (T_e/m_e)^{1/2}$ are the electron-ion relative drift speed, and the ion and electron thermal speeds, respectively. Second, the electrons behave as if they had an effective mass that is much larger than m_e and can, in fact, be comparable to the ion mass for the most important waves. Our study indicates that this instability can be used to provide strong plasma heating and is attractive for explaining effects seen in a number of experiments. The main characteristics of the instability are as follows: (1) The characteristic frequency and growth rate are comparable to the lower-hybrid frequency. [In this paper we will be concerned with time and length scales such that the ions are effectively unmagnetized. Thus, by lower-

hybrid frequency, we will mean $\omega_{pi}(1 + \omega_{pe}^2/\Omega_e^2)^{-1/2}$, where ω_{pi} , ω_{pe} , and Ω_e are the ion-plasma frequency, electron-plasma frequency, and electron cyclotron frequency.] (2) The instability is a fluid-like, nonresonant instability where nonlinear saturation grossly changes the particle distribution functions. (3) Unlike many other electron-ion instabilities which mainly heat electrons (e.g., the ordinary two-stream, the beam cyclotron, and the ion-acoustic instabilities), the modified two-stream instability results in comparable electron and ion heating. (4) The instability is insensitive to the electron-ion temperature ratio, T_e/T_i , and thus can be operative when other electrostatic instabilities cannot.

In Sec. II, we present an analysis of the electrostatic dispersion relation. In particular, we discuss the parametric dependence of the unstable roots on T_e/T_i , U/v_i , m_e/m_i , and ω_{pe}/Ω_e . We also discuss the stabilization which can occur when the electrostatic assumption is lifted and electromagnetic effects are included.

In Sec. III, we present nonlinear analyses of the instability based on quasilinear and trapping arguments. We also derive a fully nonlinear scaling law which, under certain conditions, allows us to use one computer experiment, done with a particular mass ratio (m_e/m_i), to infer results for other mass ratios. The appendix contains a related discussion of the effect of cross-field vortex-like motion caused by turbulence-induced $\mathbf{E} \times \mathbf{B}$ drifts.

Section IV presents results of one- and two-dimensional computer simulation experiments and discusses these results in relation to theory. The presence of competing microinstabilities which may coexist with the modified two-stream instability in some cases is also discussed.

Section V discusses various physical and experimental situations in which cross-field electron-ion

streaming occurs and assesses the importance of the modified two-stream instability in these cases. Among such situations are collisionless shocks, anomalous damping of large amplitude waves, momentum coupling between interpenetrating plasmas, and some turbulent heating experiments. We also discuss the possibility of inducing cross-field streaming in fusion machines, like tokamak and mirrors, for the purpose of using the modified two-stream instability for ion heating.

In Sec. VI we summarize and discuss the results of the study.

II. LINEAR THEORY

We begin by discussing the linear, electrostatic dispersion relation for homogeneous, unbounded plasmas in a constant magnetic field $\mathbf{B}_0$. Electromagnetic effects will be considered at the end of this section. The dispersion relation for electrostatic waves is $\mathbf{k} \cdot \boldsymbol{\epsilon} \cdot \mathbf{k} = 0$, where $\boldsymbol{\epsilon}$ is the plasma dielectric tensor. We rewrite this in the form

$$1 + (\mathbf{k} \cdot \mathbf{M}^e \cdot \mathbf{k} / k^2) + (\mathbf{k} \cdot \mathbf{M}^i \cdot \mathbf{k} / k^2) = 0, \quad (1)$$

where we have written $\boldsymbol{\epsilon} = \mathbb{I} + \mathbf{M}^e + \mathbf{M}^i$, $\mathbb{I}$ being the unit tensor, and $\mathbf{M}^e$, $\mathbf{M}^i$ being the electron-ion susceptibility. We consider waves for which $k\rho_i \gg 1$ and $|\omega| \gg \Omega_i$, where ρ_i is the ion Larmor radius and Ω_i is the ion cyclotron frequency, so that the ions are effectively unmagnetized. Assuming a relative drift velocity $\mathbf{U}$ between the electrons and ions and performing the analysis in the electron frame of reference, the ion contribution to Eq. (1), is

$$\mathbf{k} \cdot \mathbf{M}^i \cdot \mathbf{k} / k^2 = -(2k^2\lambda_D^2)^{-1} (T_e/T_i) Z'[(\omega - \mathbf{k} \cdot \mathbf{U}) / \sqrt{2}k v_i]. \quad (2)$$

In Eq. (2), λ_D is the electron Debye length, $Z(w)$ is the plasma dispersion function, and $Z' = dZ(w)/dw$. Taking $\mathbf{k} = (k_x, 0, k_z)$ and $\mathbf{B}_0 = (0, 0, B_0)$, so that there is a component of $\mathbf{k}$ parallel to $\mathbf{B}_0$, the electron contribution to Eq. (1) is

$$\mathbf{k} \cdot \mathbf{M}^e \cdot \mathbf{k} / k^2 = (k_x^2/k^2) M_{xx}^e + (k_z^2/k^2) M_{zz}^e, \quad (3)$$

where the electron susceptibility tensor elements are given by Stix.¹⁴ In the range of frequencies $\Omega_e \ll |\omega| \ll \Omega_e$, which applies for the modes of interest in this paper, $\mathbf{M}^e$ may be greatly simplified. Thus, subject only to the additional requirement $k\rho_e \ll 1$, where ρ_e is the electron Larmor radius, the dispersion relation becomes

$$1 - (2k^2\lambda_D^2)^{-1} \frac{T_e}{T_i} Z'\left(\frac{\omega - \mathbf{k} \cdot \mathbf{U}}{\sqrt{2}k v_i}\right) + \frac{[1 - e^{-\lambda} I_0(\lambda)]}{k^2 \lambda_D^2} - (2k^2\lambda_D^2)^{-1} e^{-\lambda} I_0(\lambda) Z'\left(\frac{\omega}{\sqrt{2}k v_e}\right) = 0, \quad (4)$$

where $\lambda \equiv k_x^2 \rho_e^2$, and I_0 is the zeroth-order modified Bessel function. Equation (4) forms the basis for our linear study of the modified two-stream instability. For definiteness we shall henceforth assume that $\mathbf{U}$ is in the x direction, $\mathbf{U} = (U, 0, 0)$.

At this point we discuss the fluid limit of Eq. (4) where resonant particle effects are unimportant. Modes for which $k\rho_e \ll 1$, $k v_i < |\omega - \mathbf{k} \cdot \mathbf{U}|$, and $k_z v_e < |\omega|$ are described by the following simplified dispersion relation^{4-8,10,11}:

$$1 + \frac{k_x^2}{k^2} \frac{\omega_{pe}^2}{\Omega_e^2} - \frac{\omega_{pi}^2}{(\omega - \mathbf{k} \cdot \mathbf{U})^2} - \frac{k_z^2}{k^2} \frac{\omega_{pe}^2}{\omega^2} = 0. \quad (5)$$

Equation (5) is analogous in form to the well-known two-stream or Buneman¹⁸ instability dispersion relation, but is valid even for drifts which are considerably smaller than v_e . Indeed, the threshold for instability is $U \gtrsim v_i$. The term $(k_x^2/k^2)(\omega_{pe}^2/\Omega_e^2)$ is due to the adiabatic polarization drift of electrons across the magnetic field, $\mathbf{v}_a = -c\hat{\mathbf{E}}_\perp/B_0\Omega_e$. The term $(k_z^2/k^2)(\omega_{pe}^2/\omega^2)$ arises because electrons are free to accelerate under an applied force only along the magnetic field. Thus, the electrons behave as if they had an effective mass $\bar{m}_e = (k^2/k_z^2)m_e$ which can be large for $(k^2/k_z^2) \gg 1$. For the unstable roots of Eq. (5), the $\mathbf{k}$ vector angles of most interest are nearly perpendicular to $\mathbf{B}_0$ with $k_z/k \sim (m_e/m_i)^{1/2}$ implying $\bar{m}_e \sim m_i$. In this case, $\text{Re}(\omega) \sim \text{Im}(\omega) \sim \omega_{LH}$ where $\omega_{LH} = \omega_{pi}/(1 + \omega_{pe}^2/\Omega_e^2)^{1/2}$ is the lower-hybrid frequency. Since $\mathbf{k}$ is nearly perpendicular to $\mathbf{B}_0$, we expect ion heating primarily in this perpendicular direction. However, since electrons are free to accelerate only along $\mathbf{B}_0$, electrons will be heated in the direction parallel to $\mathbf{B}_0$. This suggests that waves with $k_z/k \sim (m_e/m_i)^{1/2}$ should cause a perpendicular ion heating rate comparable to the parallel electron heating rate, since at these $\mathbf{k}$ vector angles the ion mass and the effective electron mass are comparable. By way of illustration, Eq. (5) can be solved exactly at $\sin\theta \equiv (k_z/k) = (m_e/m_i)^{1/2}$. We refer to this value of $\theta \equiv \theta_0$ as the equal effective mass angle because here $\bar{m}_e = m_i$. Writing $\omega' = \omega + \frac{1}{2}(\mathbf{k} \cdot \mathbf{U})$ and $\bar{m}_e = m_i$, Eq. (5) becomes a quadratic in ω'^2 from whose solution we obtain the following maximum growth rate $\gamma_{\max}$, $\gamma_{\max} = \frac{1}{2}\omega_{LH}$, which occurs at the wavenumber $k_{\max} = \sqrt{3}\omega_{LH}/U$, and real frequency $\text{Re}(\omega)_{\max} = k_{\max}U/2 = \frac{1}{2}\sqrt{3}\omega_{LH}$. This discussion of the fluid limit for $\theta = \theta_0$ is given only as an example. A more accurate prediction of the actual properties of this unstable mode depends on exact solutions of Eq. (4). However, as will become clear, an analysis of the problem at θ_0 gives good qualitative predictions of the plasma behavior.

We note that since $\omega \sim kU/2$, the condition necessary to avoid the stabilizing effects of ion Landau damping $|\omega - \mathbf{k} \cdot \mathbf{U}| > k v_i$, is roughly $U \gtrsim v_i$, while the condition necessary to avoid electron Landau resonance, $k_z v_e < \omega$ becomes a condition on the angle of wave propagation, that is, $k_z/k \lesssim U/v_e$, or $(m_i/m_e) \tan\theta \lesssim U/(T_e/m_i)^{1/2}$.

For the purposes of doing a parametric study of the linear dispersion characteristics of the instability, we rewrite Eq. (4) as follows:

$$1 + (\kappa^2)^{-1} \left(\frac{\omega_{pe}}{\Omega_e} \right)^2 [1 - \exp(-\kappa^2 \cos^2 \theta) I_0(\kappa^2 \cos^2 \theta)] - (2\kappa^2)^{-1} \left(\frac{\omega_{pe}}{\Omega_e} \right)^2 \frac{T_e}{T_i} Z' \left[\sqrt{2}^{-1} \left\{ \left(\frac{\omega}{\omega_{pi}} \right) \kappa^{-1} \left(\frac{\omega_{pe}}{\Omega_e} \right) \left(\frac{T_e}{T_i} \right)^{1/2} - U \frac{\cos \theta}{v_i} \right\} \right] - (2\kappa^2)^{-1} \left(\frac{\omega_{pe}}{\Omega_e} \right)^2 \exp(-\kappa^2 \cos^2 \theta) I_0(\kappa^2 \cos^2 \theta) \times Z' \left[\sqrt{2}^{-1} \left(\frac{\omega}{\omega_{pi}} \right) \left(\frac{m_e}{m_i} \right)^{1/2} \frac{1}{\sin \theta} \kappa^{-1} \left(\frac{\omega_{pe}}{\Omega_e} \right) \right] = 0, \quad (6)$$

where $\kappa = k\rho_e$. It is evident from Eq. (6) that four plasma parameters enter: m_e/m_i , ω_{pe}/Ω_e , T_e/T_i , and U/v_i . We wish to study the dependence of the unstable solutions of Eq. (6) on these four parameters. Equation (6) possesses a very important scaling which enables us to eliminate the parameter m_e/m_i . For small θ , which is the case of most interest as shown by numerical solutions of the dispersion relation, θ appears to lowest order only in the last term of Eq. (6) in the argument of the Z function. Thus, defining a new variable $\bar{\theta} = \theta(m_i/m_e)^{1/2}$ the explicit mass ratio dependence disappears from Eq. (6) and the roots ω/ω_{pi} vs κ at any $\bar{\theta}$ given Ω_e/ω_{pe} , T_e/T_i , and U/v_i , are independent of the mass ratio. Yet another scaling of the dispersion relation occurs in the important limit of small θ and $\kappa^2 \ll 1$. Defining

$$\bar{\omega} = (\omega/\omega_{pi}) [1 + (\omega_{pe}^2/\Omega_e^2)]^{1/2}, \\ \bar{\kappa} = \kappa(\Omega_e/\omega_{pe}) [1 + (\omega_{pe}^2/\Omega_e^2)]^{1/2},$$

Eq. (6) can be rewritten as

$$1 - \frac{T_e}{T_i} (2\bar{\kappa}^2)^{-1} Z' \left[\left(\frac{T_e}{T_i} \right)^{1/2} \frac{\bar{\omega}}{\sqrt{2}} \bar{\kappa}^{-1} - \frac{U}{\sqrt{2}v_i} \right] - (2\bar{\kappa}^2)^{-1} Z' \left(\frac{\bar{\omega}}{\sqrt{2}} \bar{\kappa}^{-1} \bar{\theta}^{-1} \right) = 0. \quad (7)$$

Thus for any $\bar{\theta}$ and given T_e/T_i and U/v_i , the solutions $\bar{\omega}$ vs $\bar{\kappa}$ are independent of mass ratio and ω_{pe}/Ω_e . Thus, when $\kappa^2 \ll 1$ and $\theta \ll 1$ it is possible to reduce the four-dimensional parameter space of possible plasma conditions to a two-dimensional space involving only U/v_i and T_e/T_i . We have tested these approximate scaling laws by solving Eq. (6) numerically. The solid curve in Fig. 1 shows a plot of the maximum growth rate normalized to the lower-hybrid frequency against $\theta(m_i/m_e)^{1/2}$ for a hydrogen plasma with $\omega_{pe}/\Omega_e = 1$, $T_e/T_i = 1$, and $U/v_i = 10$. A similar calculation for argon (atomic weight 40) gives a virtually identical curve verifying the mass ratio scaling. In order to test the ω_{pe}/Ω_e scaling, we have calculated the maximum growth rate for the same parameters

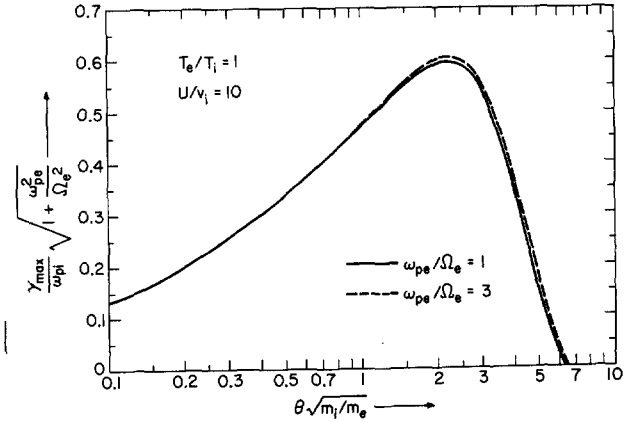


FIG. 1. Plot of the maximum growth rate (maximized over k) normalized to the lower-hybrid frequency versus $\theta(m_i/m_e)^{1/2}$ for a hydrogen plasma with two different values of ω_{pe}/Ω_e .

as above but with $\omega_{pe}/\Omega_e = 3$. This is shown in the dashed curve in Fig. 1. We note that the two curves are almost identical for $\theta(m_i/m_e)^{1/2} < 1$ and that even for larger angles there is very little difference (less than $\sim 1.5\%$ of the maximum growth rate). The accuracy of our approximate scaling law is expected since $k^2\rho_e^2 = 0.04$ at the largest $\gamma_{\max}$.

We now discuss the dependence of the instability on the two remaining plasma parameters U/v_i and T_e/T_i . Figure 2 shows plots of $\gamma_{\max}$ vs $\theta(m_i/m_e)^{1/2}$ for hydrogen for different values of U/v_i , with $\omega_{pe}/\Omega_e = 1$ and $T_e/T_i = 1$. For small angles [$\theta(m_i/m_e)^{1/2} \lesssim 0.3$] the curves are similar and agree well with the hydrodynamic dispersion relation, Eq. (5), which is independent of U/v_i . However, as θ increases, ion and electron thermal effects become more important [i.e., both $(U - \omega/k)$ and ω/k_z decrease]. When $(U - \omega/k) \sim v_i$ or $\omega/k_z \sim v_e$, we expect the wave to stabilize ($T_e = T_i$) and this should happen at smaller angles for smaller values of U/v_i . Thus, from Fig. 2 we note that decreasing U/v_i decreases the largest value of $\gamma_{\max}$ (henceforth denoted $\gamma_{\max}^*$), decreases the angle at which $\gamma_{\max}^*$ occurs, and also decreases the angle at which the instability cuts off.

Figure 3 shows plots of $\gamma_{\max}$ vs $\theta(m_i/m_e)^{1/2}$ for hydrogen for different values of T_e/T_i , with $\omega_{pe}/\Omega_e = 1$ and $U/v_i = 10$. Again, for small angles, the curves are similar and agree well with the solutions of the hydrodynamic dispersion relation. As previously noted, electron thermal effects become severe when $\theta(m_i/m_e)^{1/2} \sim U/(T_e/m_i)^{1/2}$, so that increasing T_e/T_i with U/v_i fixed decreases this angle. Indeed, we note from Fig. 3 that increasing T_e/T_i lowers $\gamma_{\max}^*$ and the angle at which it occurs. We also note, however, that at large T_e/T_i (e.g., $T_e/T_i = 10$) the electron thermal effects have increased the value of θ at which the instability cuts off thus widening the angular range of instability. The reason for this is that, for large T_e/T_i , a type of resonant instability, akin to the usual ion-acoustic instability, appears and is un-

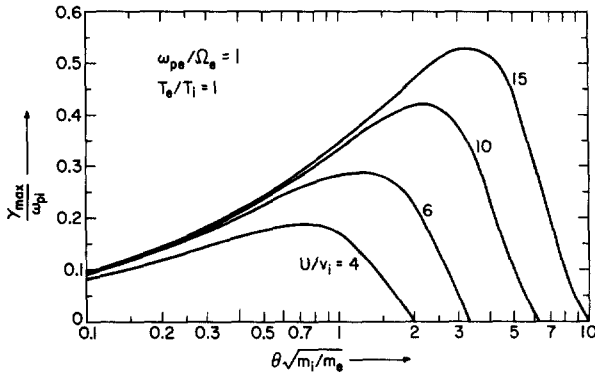


FIG. 2. Plots of the maximum growth rate normalized to ω_{pi} vs $\theta(m_i/m_e)^{1/2}$ for a hydrogen plasma for different values of U/v_i .

stable at these larger angles. This resonant-particle mode has previously been discussed.¹²

Figure 4 shows plots of $\gamma_{max}^*/\omega_{pi}$, κ^* , and ω^*/ω_{pi} vs U/v_i and T_e/T_i . In the figure κ^* and ω^* denote the values of κ and the real part of ω which give $\gamma = \gamma_{max}^*$, the growth rate maximized over both angle θ and wavenumber κ . The maximizing angle θ^* can easily be taken from earlier figures as a function of U/v_i and T_e/T_i and thus is not shown here. Of particular interest is the almost perfectly logarithmic dependence of $\gamma_{max}^*/\omega_{pi}$ on both U/v_i and T_e/T_i .

So far our discussion has been limited to $\mathbf{k}$ vectors in the $\mathbf{U}\text{-}\mathbf{B}_0$ plane [i.e., $\mathbf{k} = (k_x, 0, k_z)$ and $\mathbf{U} = (U, 0, 0)$]. To include $\mathbf{k}$ vectors with a y component as well we set $\mathbf{k} = (k_x, k_p \sin \Phi, k_p \cos \Phi)$ so that $\mathbf{k}_p$ is the component of $\mathbf{k}$ perpendicular to $\mathbf{U}$ and Φ is the angle between $\mathbf{k}_p$ and $\mathbf{B}_0$ [which was assumed to be zero subsequent to Eq. (4)]. Solutions of the dispersion relation, Eq. (4), so obtained for four values of Φ are shown in Fig. 5. In the figure the (k_{pe}, k_{xp}) plane is shown to scale for a hydrogen mass ratio with $\omega_{pe}/\Omega_e = 1$, $T_e/T_i = 1$, and $U/v_i = 10$. The outer lobe-shaped curves in each of the four subfigures surround the linearly unstable region and thus are lines of marginal stability. Increasing Φ toward 90° tends to spread out the angular range of instability by reducing $\mathbf{k} \cdot \mathbf{B}_0$ for a given value of $\mathbf{k}_p$ (here, $\mathbf{k} \cdot \mathbf{B}_0 = k_p B_0 \cos \Phi$) since the equal mass angle in the (k_p, k_x) plane is a function of Φ . In Fig. 5 the center of the lobe-shaped unstable regions lies at about

$$\mathbf{k} \cdot \mathbf{B}_0 / |\mathbf{k}| |\mathbf{B}_0| = 2.25(m_e/m_i)^{1/2}.$$

Increasing Φ also decreases the large k cutoff of the instability. For example, at $\Phi = 80^\circ$ there are no unstable waves having $k_{xp} \gtrsim 0.5$. Further increase of Φ decreases the high k cutoff even further until at $\Phi = 90^\circ$ the mode is fully stabilized since $\mathbf{k} \cdot \mathbf{B}_0$ is zero. The actual growth rates γ/ω_{pi} are also plotted in the unstable region of each of the subfigures as a function of k_{pe} for the values of $k_{xp} = 0, 0.05, 0.10, 0.15$, etc.

The scale of γ/ω_{pi} is shown for each value of k_{xp} at the right of the figure.

It is interesting to note that the curve of marginal stability approaches a straight line in each of the four subfigures to Fig. 5 for small k_{xp} . This linearity can be seen analytically from Eq. (7). When κ is small, the 1 may be neglected in which case the dispersion relation depends only on ω/κ . Thus, any root giving $\omega(\kappa)$ gives, in fact, a whole family of roots $\alpha\omega(\alpha\kappa)$, where α is a real multiplicative constant, as long as $\alpha\kappa$ is sufficiently small.

We conclude this section with a brief description of electromagnetic and finite β effects on the instability (for further details see Ref. 10). The electromagnetic dispersion relation in the fluid limit assuming $k_s \ll k_x$ and $\omega \ll \Omega_e$ is¹⁰

$$1 - [\Omega_e^2 / (\omega - \mathbf{k} \cdot \mathbf{U})^2] - (\Omega_e^2 / \omega^2) = 0, \quad (8)$$

where

$$\Omega_e^2 = \omega_{pe}^2 / \Delta, \quad \Omega_b^2 = \frac{k_x^2}{k^2} \frac{\omega_{pe}^2}{\Delta(1 + \omega_{pe}^2 / k^2 c^2)},$$

$$\Delta \equiv 1 + (\omega_{pe}^2 / \Omega_e^2) [1 + (1 + \beta_e)^{-1} \omega_{pe}^2 / k^2 c^2],$$

and β_e is the ratio of electron pressure to magnetic pressure, $\beta_e = (nT_e) / (B_0^2 / 8\pi)$. Equation (8) is analogous in form to Eq. (5) but contains extra wave-number dependences through the terms in Ω_e and Ω_b . These terms can be important depending on the size of β_e , $\omega_{pe}^2 / k^2 c^2$. A marginal stability analysis¹⁰ of Eq. (8) shows that the electromagnetic terms are stabilizing and that instability occurs for $\bar{m}_e \gtrsim m_i$, where the interaction is expected to be strongest on the ions, only if

$$U/V_A \gtrsim (1 + \beta_e)^{1/2}, \quad (9)$$

where V_A is the Alfvén speed. For larger wave-vector angles ($\bar{m}_e < m_i$) the electromagnetic stabilization condition, (9), relaxes somewhat,¹⁰ however, if the angle is too large, the wave may be stabilized by thermal effects. The electrostatic limit, Eq. (5), is strictly

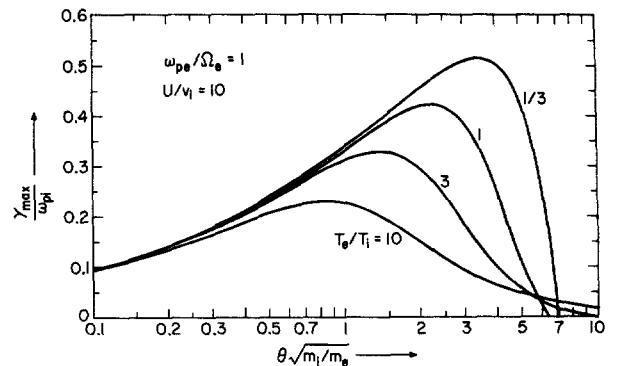


FIG. 3. Plots of the maximum growth rate normalized to ω_{pi} vs $\theta(m_i/m_e)^{1/2}$ for a hydrogen plasma for different values of T_e/T_i .

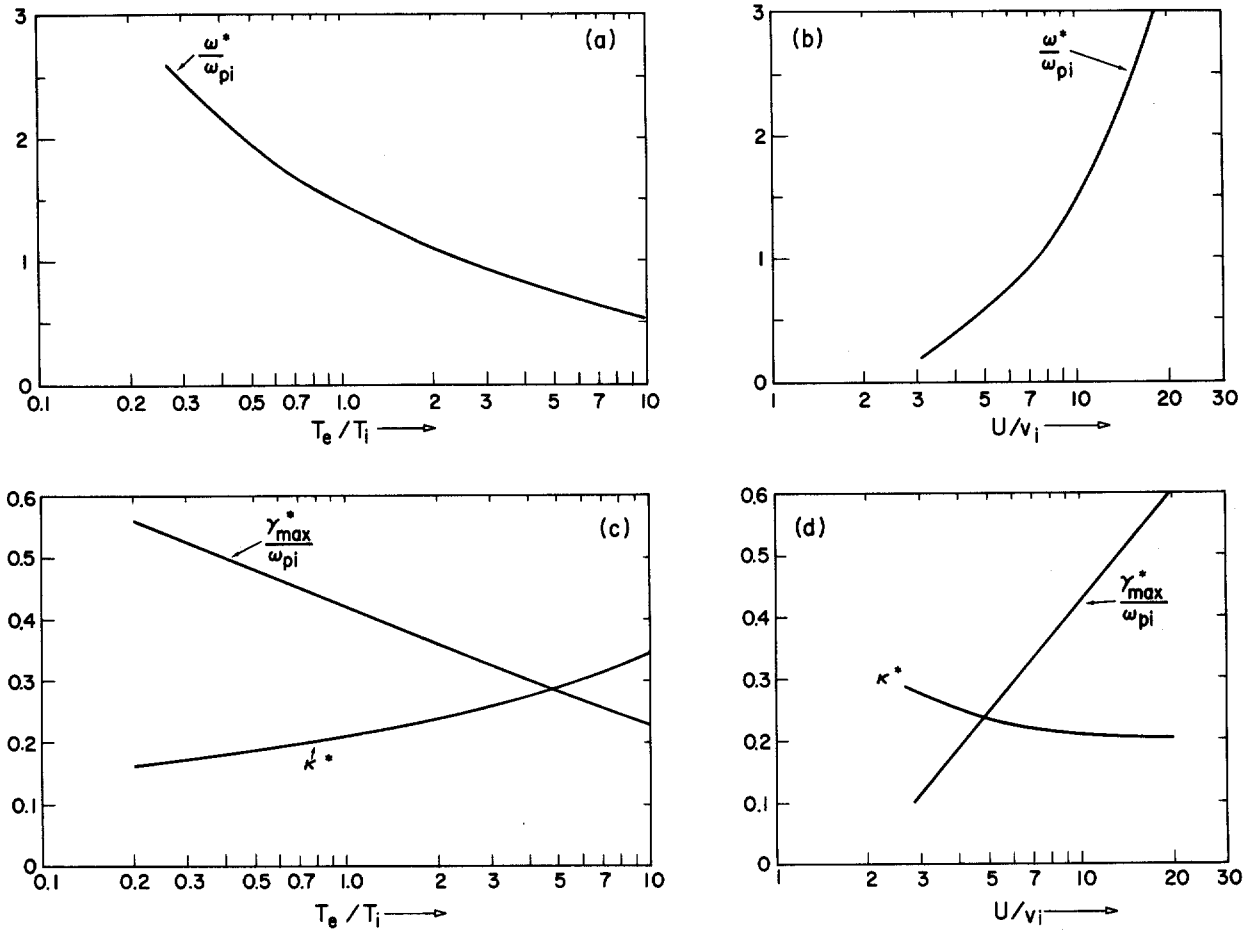


FIG. 4. Plots of $\gamma_{max}^*/\omega_{pi}$, κ^* , and ω^*/ω_{pi} vs U/v_i and T_e/T_i for a hydrogen plasma for $\omega_{pe}/\Omega_e = 1$. In subfigures (a) and (c) $U/v_i = 10$, and in subfigures (b) and (d) $T_e/T_i = 1$.

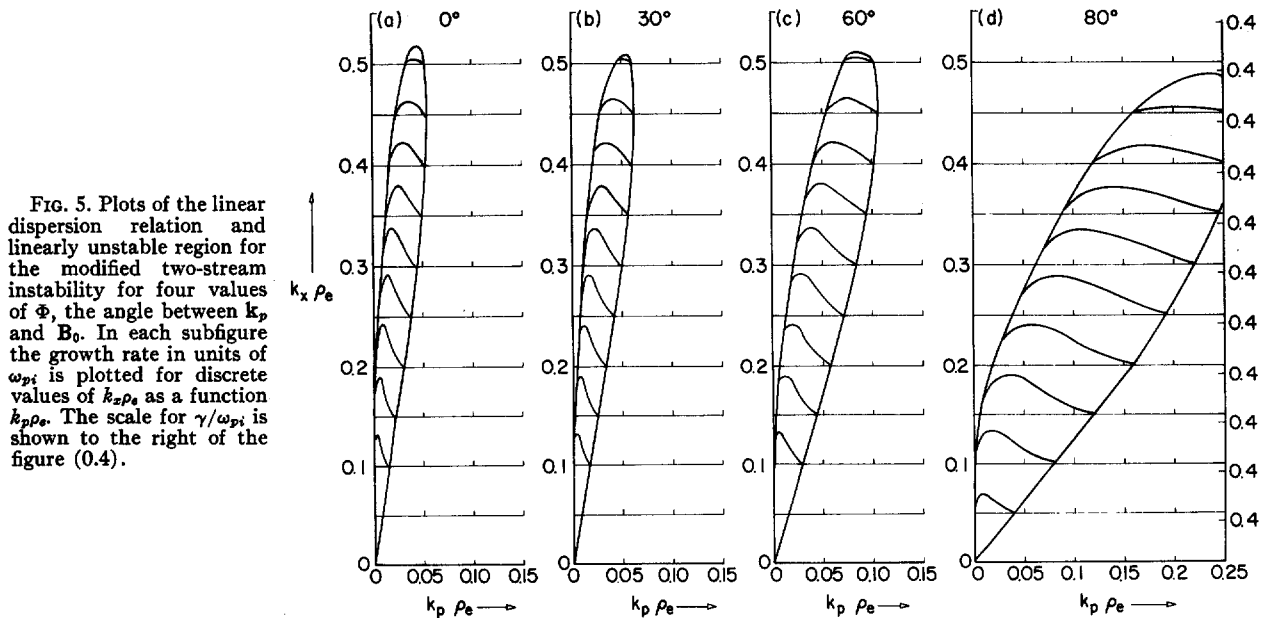


FIG. 5. Plots of the linear dispersion relation and linearly unstable region for the modified two-stream instability for four values of Φ , the angle between $\mathbf{k}_p$ and $\mathbf{B}_0$. In each subfigure the growth rate in units of ω_{pi} is plotted for discrete values of $k_x \rho_e$ as a function of $k_p \rho_e$. The scale for γ/ω_{pi} is shown to the right of the figure (0.4).

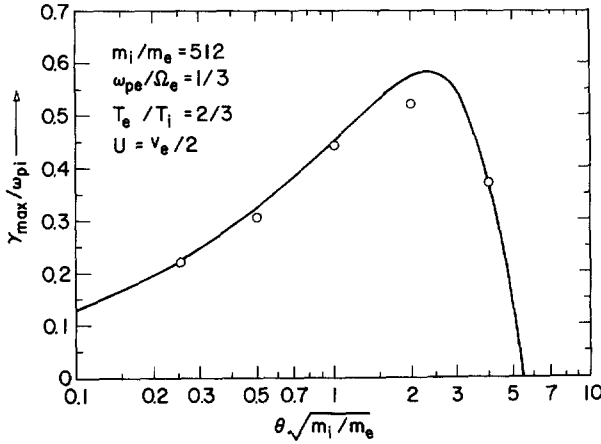


FIG. 6. Plot of the maximum growth rate normalized to ω_{pi} vs $\theta(m_i/m_e)^{1/2}$ from Eq. (4) for the indicated parameters. The circles are the measured maximum growth rates from a set of one dimensional computer simulations at different values of θ .

valid only if $U^2 \ll V_A^2(1+\beta_e)$. Since $U \lesssim v_i$, Eq. (9) can only be satisfied if $v_i < V_A(1+\beta_e)^{1/2}$ or $\beta_i < 2(1+\beta_e)$, where β_i is $nT_i/(B_0^2/8\pi)$.

III. NONLINEAR THEORY

In order to simplify our nonlinear analysis, we shall assume that the time and length scales on which the instability operates are much shorter than the ion Larmor period and radius but much longer than the electron Larmor period and radius. Our solutions to the linear dispersion relation have already shown that these conditions are satisfied in interesting regions of parameter space. Thus, the ions can be treated as unmagnetized, and the electron motion across the magnetic field may be treated in the drift approximation. The relevant set of equations is then

$$\frac{\partial F_i}{\partial t} + \mathbf{v} \cdot \nabla F_i - \frac{e}{m_i} \nabla \phi \cdot \frac{\partial F_i}{\partial \mathbf{v}} = 0, \quad (10)$$

$$\frac{\partial F_e}{\partial t} + v_z \frac{\partial F_e}{\partial z} + \frac{e}{m_e} \frac{\partial \phi}{\partial z} \frac{\partial F_e}{\partial v_z} + \nabla \cdot (\mathbf{v}_D F_e) = 0, \quad (11)$$

$$\nabla^2 \phi = -4\pi e (\int F_i d\mathbf{v} - \int F_e d\mathbf{v}_z), \quad (12)$$

where F_i is the ion distribution function, $F_e(v_z, x, y, z)$ is the distribution function of electron guiding centers, ϕ is the electric potential, and $\mathbf{v}_D$ is the cross-field electron drift which consists of an $\mathbf{E} \times \mathbf{B}$ drift and a polarization drift

$$\mathbf{v}_D(v_z, x, y, z) = c(\mathbf{E} \times \mathbf{B}_0/B_0^2) + \mathbf{v}_p, \quad (13)$$

and the polarization drift is given by $\mathbf{v}_p = -c(\Omega_e B_0)^{-1} \times d\mathbf{E}_\perp/dt$, or

$$\mathbf{v}_p = -\frac{c}{\Omega_e B_0} \left(\frac{\partial \mathbf{E}_\perp}{\partial t} + \mathbf{v}_D \cdot \Delta \mathbf{E}_\perp + v_z \frac{\partial \mathbf{E}_\perp}{\partial z} \right). \quad (14)$$

We will discuss an important mass ratio scaling of these nonlinear equations in Sec. IIIC.

A. Quasilinear Theory

Applying the standard quasilinear procedure to the above set of equations, we obtain quasilinear diffusion equations for the spatially averaged distribution functions $\bar{F}_i$ and $\bar{F}_e$ and for the average electron kinetic energy in cross-field drifts which we denote $K_\perp^e$,

$$\frac{\partial \bar{F}_i}{\partial t} = \frac{\partial}{\partial \mathbf{v}} \cdot \mathbf{D}^i \cdot \frac{\partial \bar{F}_i}{\partial \mathbf{v}},$$

$$\mathbf{D}^i = \frac{e^2}{m_i^2} \sum_{\mathbf{k}} \frac{\mathbf{k} \mathbf{k} |\phi(\mathbf{k})|^2 \gamma_{\mathbf{k}}}{(\Omega_{\mathbf{k}} - \mathbf{k} \cdot \mathbf{v})^2 + \gamma_{\mathbf{k}}^2}, \quad (15)$$

$$\frac{\partial \bar{F}_e}{\partial t} = \frac{\partial}{\partial v_z} D^e \frac{\partial \bar{F}_e}{\partial v_z},$$

$$D^e = \frac{e^2}{m_e^2} \sum_{\mathbf{k}} \frac{k_z^2 |\phi(\mathbf{k})|^2 \gamma_{\mathbf{k}}}{(\Omega_{\mathbf{k}} - k_z v_z)^2 + \gamma_{\mathbf{k}}^2}, \quad (16)$$

$$\frac{dK_\perp^e}{dt} = \frac{\omega_{pe}^2}{\Omega_e^2} \frac{d}{dt} \sum_{\mathbf{k}} |\mathbf{E}_{\mathbf{k}}|^2 / 8\pi, \quad (17)$$

where the solution of the dispersion relation is denoted $\omega_{\mathbf{k}} = \Omega_{\mathbf{k}} + i\gamma_{\mathbf{k}}$. Since the waves are primarily perpendicular to $\mathbf{B}_0$ [for the maximum growth rate our solutions to the linear dispersion relation, Figs. 1 to 5, imply $|\mathbf{k}_\perp| \sim (m_i/m_e)^{1/2} k_z$], we may, therefore, approximate $\mathbf{k} \mathbf{k}$ in Eq. (15) by $\mathbf{k}_\perp \mathbf{k}_\perp$. It is thus clear from Eq. (15) that the ion heating is primarily perpendicular to $\mathbf{B}_0$, while Eq. (16) implies electron heating along $\mathbf{B}_0$. The energy change represented in Eq. (17) must be thought of as a type of "reversible" heating since it merely represents the kinetic energy in $\mathbf{E} \times \mathbf{B}$ drifts associated with the wave motion. Although ion gyromotion is on a time scale much longer than the linear wave growth rate, for long times ($t \gtrsim 2\pi\Omega_i^{-1}$), Eq. (15) must be replaced by

$$\frac{\partial \bar{F}_i}{\partial t} + \Omega_i (\mathbf{v} \times \mathbf{z}_0) \cdot \frac{\partial \bar{F}_i}{\partial \mathbf{v}} = \frac{\partial}{\partial \mathbf{v}} \cdot \mathbf{D} \cdot \frac{\partial \bar{F}_i}{\partial \mathbf{v}}. \quad (18)$$

The effect of the ion gyroterm in conjunction with the diffusion operator $\mathbf{D}$ in Eq. (18) is to isotropize $\bar{F}_i$ in the plane perpendicular to $\mathbf{B}_0$ in a few ion gyrotimes. In this paper we consider times $t < 2\pi\Omega_i^{-1}$.

It is of interest to investigate the quasilinear equations in the hydrodynamic limit [i.e., the same limit in which Eq. (5) applies]. In this case, $\mathbf{v}$ in Eq. (15) is replaced by $\mathbf{U}_i$ and v_z in Eq. (16) is replaced by U_e , where $\mathbf{U}_i$ is the ion drift, $\mathbf{U}_i = (\int \bar{F}_i \mathbf{v} d\mathbf{v}) / (\int \bar{F}_i d\mathbf{v})$, and U_e is the electron drift along $\mathbf{B}_0$. For example, at the equal effective mass angle, (with $k_y = U_e = 0$, $\mathbf{U}_i = U \mathbf{x}_0$) the maximum growth rate is $\gamma_{\mathbf{k}} = \omega_{pi} / (2\xi)$ [where $\xi \equiv (1 + \omega_{pe}^2 / \Omega_e^2)^{1/2}$] and occurs at $\Omega_{\mathbf{k}} = \sqrt{3} \gamma_{\mathbf{k}}$ and $k_z \approx 2\Omega_{\mathbf{k}} / U$. Putting this in Eqs. (15) and (16)

and taking moments we obtain

$$\frac{dK_z^i}{dt} = \xi^2 \frac{dW}{dt}, \quad (19)$$

$$\frac{dK_z^e}{dt} = \xi^2 \frac{dW}{dt}, \quad (20)$$

$$\frac{d}{dt} (\frac{1}{2} m_i n_0 U^2) = -3\xi^2 \frac{dW}{dt}, \quad (21)$$

where $W(t)$ is the total electric field energy, $W(t) = \sum_k |E_k|^2 / 8\pi$, and K_z^e and K_z^i are the parallel electron and perpendicular ion kinetic energies in the rest frame of that species. These equations, (19)–(21) will be compared with simulation results later. Note that in the hydrodynamic limit the electron and ion heating, Eqs. (19) and (20), is purely nonresonant heating (i.e., reversible).

B. Trapping

The results of our computer simulations of the modified two-stream instability indicate that nonlinear stabilization is due to particle trapping in the potential troughs of the waves. In this subsection we estimate the effects of trapping on the plasma. In the electron frame of reference, the wave potential energy required to begin trapping the ions is

$$e\phi_i = \frac{1}{4} m_i [U - (\omega/k)]^2, \quad (22)$$

where $U - \omega/k$ defines the trapping width for ions at saturation. Since the electrons are tied to the magnetic field lines and thus can move freely only along $\mathbf{B}_0$, the wave potential energy necessary to begin trapping them is given in terms of their phase velocity parallel to $\mathbf{B}_0$: $e\phi_e = \frac{1}{4} m_e (\omega/k_z)^2 = \frac{1}{4} \bar{m}_e (\omega/k)^2$. Which species begins to trap first depends on

$$\frac{\phi_i}{\phi_e} = \frac{m_i}{\bar{m}_e} \frac{(U - \omega/k)^2}{(\omega/k)^2}. \quad (23)$$

If $\phi_i/\phi_e > 1$, the electrons begin to trap first while if $\phi_i/\phi_e < 1$, the ions begin to trap first. Numerical solutions of the linear dispersion relation show that $\phi_i/\phi_e \approx 1$ for the values of $m_i/\bar{m}_e$ of most interest [e.g., for the parameters of Fig. 1 (solid curve), $\phi_i = \phi_e$ at $\theta = \theta_0$ and $\phi_i = (0.93)\phi_e$ at $\theta = 2.25\theta_0$ which corresponds to the fastest growing wave]. Thus, based on Eq. (23) both species begin to trap at about the same time.

The other consideration is the trapping or "bounce" time τ_B . For the ions, the trapping time is

$$\tau_{B_i} = (m_i / ek^2 \phi)^{1/2},$$

while for the electrons the trapping time is

$$\tau_{B_e} = (\bar{m}_e / ek^2 \phi)^{1/2}.$$

Thus,

$$\tau_{B_i} / \tau_{B_e} \approx (m_i / \bar{m}_e)^{1/2}, \quad (24)$$

so that for $\bar{m}_e > m_i$, the ions should bounce in the wave before the electrons and saturate the instability. For $\bar{m}_e < m_i$, on the other hand, the electrons should bounce first and cause stabilization of the exponential wave growth. Since $\phi_i \approx \phi_e$, however, we expect that eventually (i.e., after a time of order τ_{B_i} , or τ_{B_e} , whichever is longer) both species will acquire final temperatures which are comparable. This result, borne out by the simulations, will be true if the electrostatic energy in the waves does not dissipate on a time of the order of the trapping time. An estimate of the ratio of electrostatic wave energy to plasma thermal energy at saturation can easily be obtained from Eq. (22). The result is

$$\frac{W}{nT_{e||}} = \frac{k^2 \lambda_{D||}^2}{16} \frac{(U - \omega/k)^4}{(T_{e||}/m_i)^2}, \quad (25)$$

where W is the electrostatic energy density, $T_{e||}$ is the electron temperature parallel to $\mathbf{B}_0$, and ω and k correspond to the fastest growing mode.

In particular for $\theta = \theta_0$, $\phi_i = \phi_e$, and $\tau_{B_i} = \tau_{B_e}$, and we expect the electrons and ions to heat equally parallel to $\mathbf{B}_0$ and perpendicular to $\mathbf{B}_0$, respectively. Further, at this value of θ the saturation trapping width in velocity space V_T is the same for both ions and electrons with effective mass $\bar{m}_e = m_i$, $V_T = \frac{1}{2}U$, since the wave phase speed is $\frac{1}{2}U$. Thus, based on the consideration of waves at θ_0 we expect the final state to be one in which $(T_{i\perp}/m_i)^{1/2} = (T_{e||}/m_i)^{1/2} \sim \frac{1}{2}U$, where U is the initial drift speed. This is confirmed by a one-dimensional computer simulation. Using this final state and replacing k , and ω/k in Eq. (25) with $kU = \sqrt{3}\omega_{pi}/(1 + \omega_{pe}^2/\Omega_e^2)^{1/2}$, and $U/2$, respectively, corresponding to the fastest growing mode at $\theta = \theta_0$, we obtain

$$W/nT_{e||} \sim 0.05(1 + \omega_{pe}^2/\Omega_e^2)^{-1}. \quad (26)$$

This agrees quite well with the computer simulation results discussed in Sec. IV. We reiterate that the numerical solutions to the linear dispersion relation, Eq. (4), indicate that the most important waves have $\theta \sim \theta_0$, so that we can expect sizeable transfer of drift energy into thermal energy in real plasmas resulting in $T_i \sim T_e$. This observation is confirmed by the results of our two-dimensional simulations.

C. Mass Ratio Scaling

In the discussion of the linear dispersion relation, we saw that for wave-vector angles $\theta \ll 1$ it is possible to eliminate the mass ratio dependence of the dispersion relation by introducing a scaled angular variable $\bar{\theta} = (m_i/m_e)^{1/2}\theta$. In this subsection we demonstrate that, under certain conditions, this same result applies to the fully nonlinear problem. This is an important point since it implies that a complete computer simulation at one mass ratio can be scaled to obtain results at any other mass ratio. We assume (1) that varia-

tions are permitted only in the x - z plane with $\mathbf{U} = U\mathbf{x}_0$ and $\mathbf{B}_0 = B_0\mathbf{z}_0$, (2) that variations in the x direction are much faster than in the z direction by a factor of order $(m_i/m_e)^{1/2}$, and (3) that $k\rho_e \ll 1$, $k\rho_i \gg 1$, $\Omega_i \ll \omega \ll \Omega_e$. The second assumption implies that x derivatives may be assumed much larger than z derivatives in the Vlasov-Poisson equations and the third assumption allows us to treat the ions as unmagnetized and to use the drift kinetic equations for the electrons. Assumption (2) seems to be well justified on the basis of the numerical solutions of the dispersion relation; the assumption $k\rho_e \ll 1$ is probably the most restrictive. Introducing the following dimensionless variables, $\bar{t} = \omega_{pi}t$, $\bar{x} = \omega_{pi}x/U$, $\bar{\mathbf{v}}_p \equiv \mathbf{v}_p/U \equiv \bar{v}_p\mathbf{x}_0$, $\bar{F}_{e,i} = F_{e,i}/n_{e,i}$, $\bar{\phi} = e\phi/(m_i U^2)$, $\bar{z} = (m_e/m_i)^{1/2} \omega_{pi}z/U$, and normalizing the ion velocity and the electron v_z to their respective thermal speeds, we obtain to lowest order in $(m_e/m_i)^{1/2}$ [cf. Eqs. (10)-(14)]:

$$\frac{\partial \bar{F}_i}{\partial \bar{t}} + \left(\frac{v_i}{U}\right) \bar{v}_x \frac{\partial \bar{F}_i}{\partial \bar{x}} - \left(\frac{U}{v_i}\right) \frac{\partial \bar{\phi}}{\partial \bar{x}} \frac{\partial \bar{F}_i}{\partial \bar{z}} = 0, \quad (27)$$

$$\frac{\partial \bar{F}_e}{\partial \bar{t}} + \bar{v}_z \left(\frac{v_i}{U}\right) \left(\frac{T_e}{T_i}\right)^{1/2} \frac{\partial \bar{F}_e}{\partial \bar{z}} + \left(\frac{U}{v_i}\right) \left(\frac{T_i}{T_e}\right)^{1/2} \frac{\partial \bar{\phi}}{\partial \bar{z}} \frac{\partial \bar{F}_e}{\partial \bar{x}} + \frac{\partial}{\partial \bar{x}} \bar{v}_p \bar{F}_e = 0, \quad (28)$$

$$\bar{v}_p = \frac{\omega_{pe}^2}{\Omega_e^2} \left\{ \frac{\partial}{\partial \bar{t}} + \bar{v}_p \frac{\partial}{\partial \bar{x}} + \left(\frac{v_i}{U}\right) \left(\frac{T_e}{T_i}\right)^{1/2} \bar{v}_z \frac{\partial}{\partial \bar{z}} \right\} \frac{\partial \bar{\phi}}{\partial \bar{x}}, \quad (29)$$

$$\frac{\partial^2 \bar{\phi}}{\partial \bar{x}^2} = \int \bar{F}_e d\bar{v}_z - \int \bar{F}_i d\bar{v}. \quad (30)$$

Thus, using the above normalizations, the only dimensionless variables which appear in the fully nonlinear equations are U/v_i , T_e/T_i , and ω_{pe}/Ω_e ; $(m_e/m_i)^{1/2}$ does not appear.

If variations in y were included, then new terms would appear in Eqs. (28) and (29) due to $\mathbf{E} \times \mathbf{B}$ drifts [cf., Eqs. (11), (13), and (14)]. These terms do not scale with mass ratio in the way just described. However, since most of our simulations have been done in the $\mathbf{U}$ - $\mathbf{B}_0$ plane, this scaling is still useful. In the appendix, we discuss the affect of including this nonlinearity due to the convection of electrons by $\mathbf{E} \times \mathbf{B}$ drifts.

IV. COMPUTER SIMULATION EXPERIMENTS

The one- and two-dimensional electrostatic plasma simulations reported here were performed on the IBM 360/91 computer using the P POWER simulation code developed at the Naval Research Laboratory.¹⁵ This program uses the discretization interpolation within cell particle interpolation technique¹⁵ and the finite-sized particle approximation of Dawson *et al.*¹⁶ to reduce numerical fluctuations in the simulations. Previous tests with this plasma simulation program¹⁵

allow us to conclude that numerically enhanced binary collisions have a negligible effect on the calculations reported here. All simulations discussed in this paper were performed using periodic boundary conditions with $\mathbf{E}_k = 0$ for $k=0$ and with unmagnetized ions in the initial rest frame of the magnetized electrons. The time step was $\delta t = 0.2\omega_{pe}^{-1}$. The parameters for most of the simulations were chosen so as to exclude other known electrostatic instabilities. Assuming $U < v_e$ eliminates classical electron-ion two-stream instabilities of the Buneman type,¹³ $T_i \gtrsim T_e$ renders ordinary ion-acoustic waves stable due to ion Landau damping.¹⁷ The condition $\Omega_e > \omega_{pe}$ eliminates the beam cyclotron instability^{1,2} which requires $(\Omega_e/\omega_{pe})(v_e/U) < 1$ for appreciable growth. Thus, for the above parameters the modified two-stream instability should be the strongest electrostatic instability operative when a spatially homogeneous description is adequate. We give a further discussion of the effect of competing electrostatic microinstabilities at the end of this section.

We begin by presenting the results of a set of one-dimensional simulations with mass ratio $m_i/m_e = 512$ designed to test the theory presented in the preceding

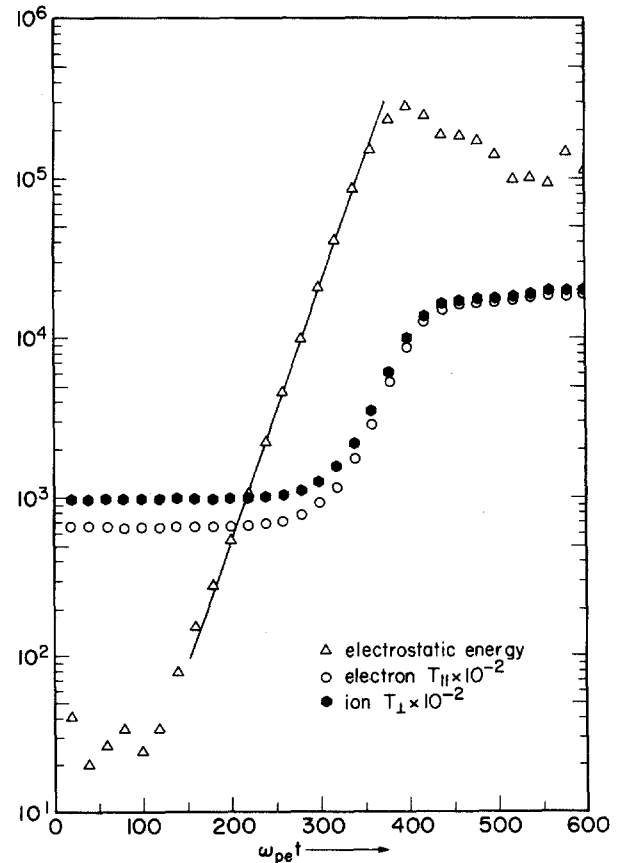


FIG. 7. Time evolution of electrostatic field energy ($\Sigma_k |E_k|^2/8\pi$), parallel electron temperature and perpendicular ion temperature for the one-dimensional computer simulation at $\theta = \theta_0$ with the parameters of Fig. 6. The straight line is drawn to emphasize the exponential development.

TABLE I. The first five columns refer to the changes in the x , y , and z components of thermal energy, the drift energy, and the field energy relative to the initial drift energy of the ions (which is nearly equal to the entire system energy). All changes Δ refer to the values of the indicated quantities at the end of the experiment ($\omega_{pe}t=600$) minus the initial value.

Species	$\Delta(X \text{ Therm } En)$ $\frac{1}{2}m_i U_0^2$	$\Delta(Y \text{ Therm } En)$ $\frac{1}{2}m_i U_0^2$	$\Delta(Z \text{ Therm } En)$ $\frac{1}{2}m_i U_0^2$	$\Delta(\text{Drift } En)$ $\frac{1}{2}m_i U_0^2$	$\Delta(\text{Field } En)$ $\frac{1}{2}m_i U_0^2$	Final thermal En Final ion drift En
Electrons	0.00	0.21	0.00	0.11	...	0.56
Ions	0.23	0.00	0.00	-0.56	...	0.63
Fields	...	...	...	...	0.01	...

sections as a function of angle θ . These calculations were performed on a system 256 cells long with 2^{14} ions and 2^{14} electrons, with initial parameters $\Omega_e = 3\omega_{pe}$, $T_i/T_e = 1.5$, $U/v_e = 0.5$, and $\lambda_D = 1$ cell. The magnetic field was chosen such that $B_z = 0$, $B_x/B_y = \tan\theta$, and the simulation was done for five different values of θ . The results of the linear stage of development are summarized in Fig. 6, where we have plotted the maximum growth rate scaled to ω_{pi} vs $\theta(m_i/m_e)^{1/2}$ based on the dispersion relation of Eq. (4). The circles are computer "experimental" growth rates for various values of θ . Each refers to the fastest growing mode observed in that simulation for fixed θ . Note that the agreement between linear theory and simulation is excellent. As an example of the full nonlinear development of the modified two-stream instability, we now discuss in some detail the particular simula-

tion of this set performed at the equal effective mass angle, $\theta_0 = (m_e/m_i)^{1/2}$. Figure 7 contains the time history of the total electrostatic energy in the unstable waves as well as the electron temperature parallel to $\mathbf{B}_0$ and the ion temperature perpendicular to $\mathbf{B}_0$. The electron heating rate along $\mathbf{B}_0$ is approximately the same as the ion heating rate across $\mathbf{B}_0$ resulting in a final electron temperature approximately equal to the final ion temperature, as predicted. The cross-field electron heating is much smaller in agreement with Eq. (17) with $\omega_{pe}^2/\Omega_e^2 = \frac{1}{3}$. Comparison with the quasilinear rate equations, Eqs. (19) and (20), shows that the initial nonlinear development of the instability can be described by quasilinear theory with the energy being partitioned to the particles and field at about the same rate. During the time interval $t = 200\omega_{pe}^{-1}$ to $t = 300\omega_{pe}^{-1}$ the observed change in parallel electron thermal energy and perpendicular ion thermal energy relative to the change in electrostatic field energy was 1.25 and 1.15, respectively, compared with

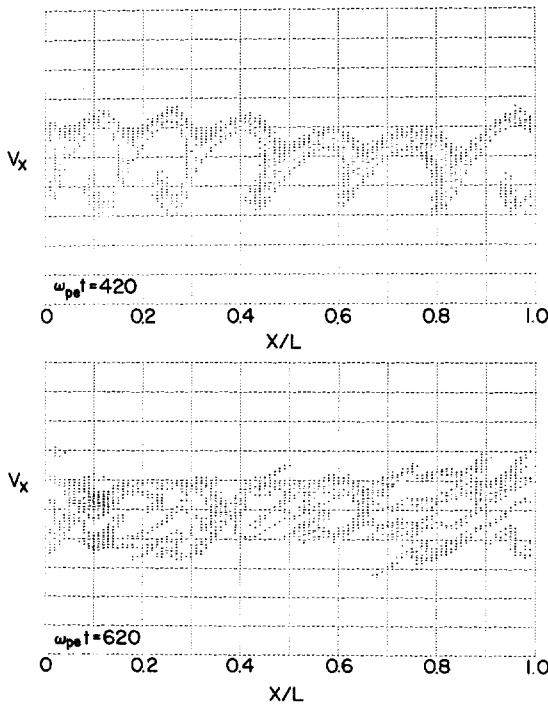


FIG. 8. Computer printouts of $x-v_x$ phase space for the ions at the end of the exponential stage ($\omega_{pe}t=420$) and at the end of the run ($\omega_{pe}t=620$). L denotes the system length and the phase space plots correspond to the simulation of Fig. 7.

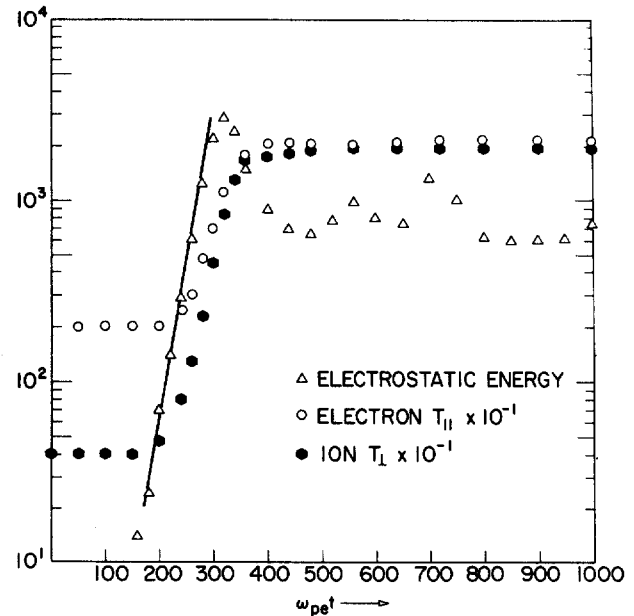


FIG. 9. Time evolution of electrostatic field energy, parallel electron temperature and perpendicular ion temperature for a one-dimensional simulation with parameters identical to those for the simulation of Fig. 7 except that $T_e = 5T_i$ initially.

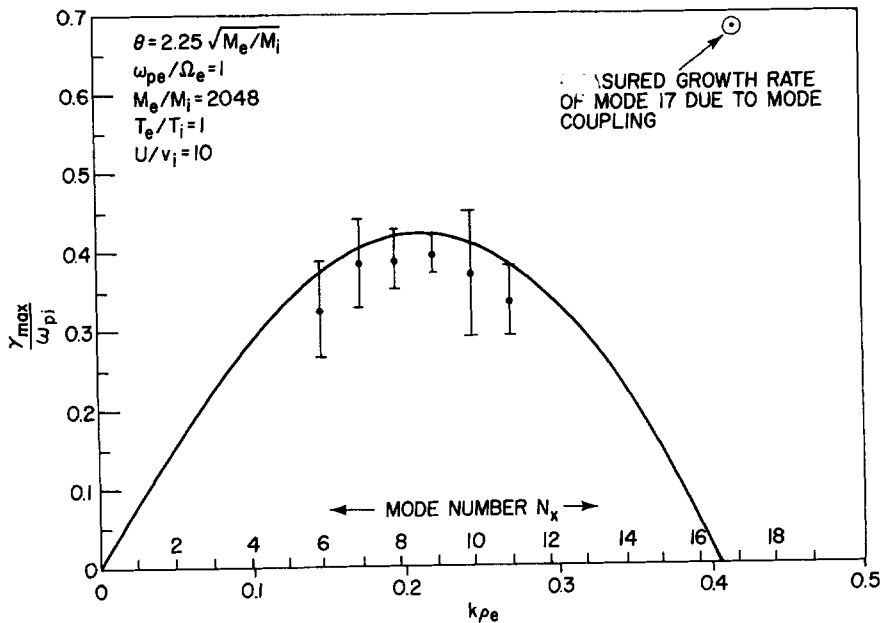


FIG. 10. Plot of maximum growth rate normalized to ω_{pi} vs $k\rho_e$ based on Eq. (4) for the indicated parameters. The dots and associated error bars refer to a corresponding one-dimensional simulation. The simulation mode numbers are also indicated on the horizontal axis.

the theoretical values of 1.11 and 1.11, respectively, predicted by Eqs. (19) and (20). However, for later times the parallel electron and perpendicular ion thermal energies increase much faster than the non-resonant quasilinear equations indicate, leading to a final thermal energy over an order of magnitude larger than the final electric field energy. This is true of all the simulations reported here. The inapplicability of nonresonant quasilinear arguments to the final heating phase is expected due to the presence of strongly nonlinear processes which lead to irreversible heating, e.g., particle trapping. Well-defined vortices are observed in plots of the x - v phase space at the end of the exponential phase of development (see Fig. 8), and we thus attribute stabilization of the instability to particle trapping. That the temperatures at the end of the run correspond to random thermal motion is also evidenced by phase space plots. At the end of the run, as seen in Fig. 8, these vortices have essentially been filled in. Table I summarizes the disposition of the free energy available in the ion streaming motion. We note the very important result that about half of the original free energy (ion energy = $\frac{1}{2}m_i U_0^2$) is converted into plasma thermal energy in several ω_{pi}^{-1} and partitioned almost equally between the electrons ($\parallel$ to $\mathbf{B}_0$) and ions ($\perp$ to $\mathbf{B}_0$). This is in good agreement with trapping theory at θ_0 , (cf. Sec. IIIB) since the saturation trapping width $(U - \omega/k)$, for the ions is $\frac{1}{2}U_0$. Thus, the final state should correspond to $(T_{i\perp}/m_i)^{1/2} \approx (T_{e\parallel}/m_e)^{1/2} \sim \frac{1}{2}U_0$ as observed. The presence of strong trapping can account for the small final ratio of electrostatic to ion drift or thermal energy. Indeed, the observed value of $W/nT_{e\parallel} \approx 0.05$ is in excellent agreement with the prediction of Eq. (26). Finally, Table I shows that 11% of the initial free ion drift energy is converted into electron-drift

energy along $\mathbf{B}_0$. This drift is an artifact of the one-dimensional simulations. In the more realistic two-dimensional runs discussed later, symmetry of the $\mathbf{k}$ vectors about the direction perpendicular to $\mathbf{B}_0$ essentially eliminates a systematic electron drift along $\mathbf{B}_0$.

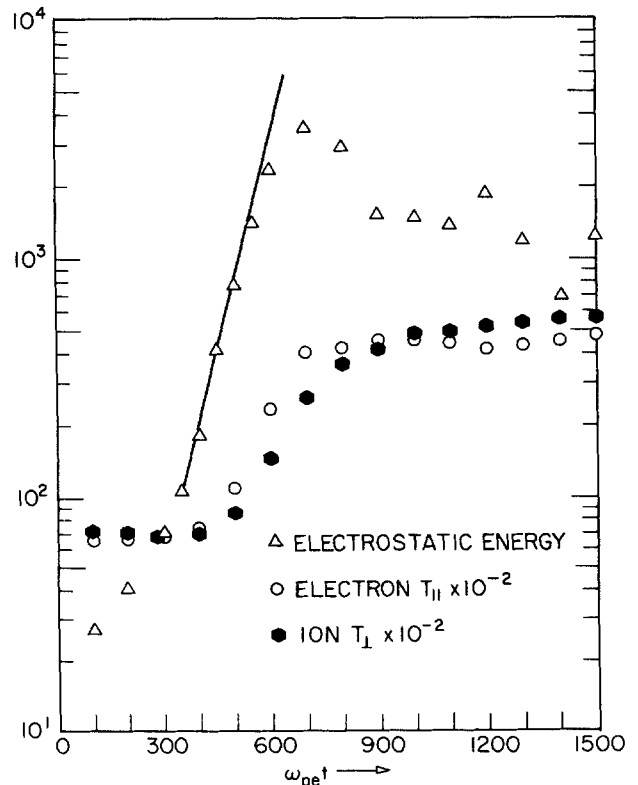


FIG. 11. Time evolution of the electrostatic field energy, parallel electron temperature, and perpendicular ion temperature for the one-dimensional simulation of Fig. 10.

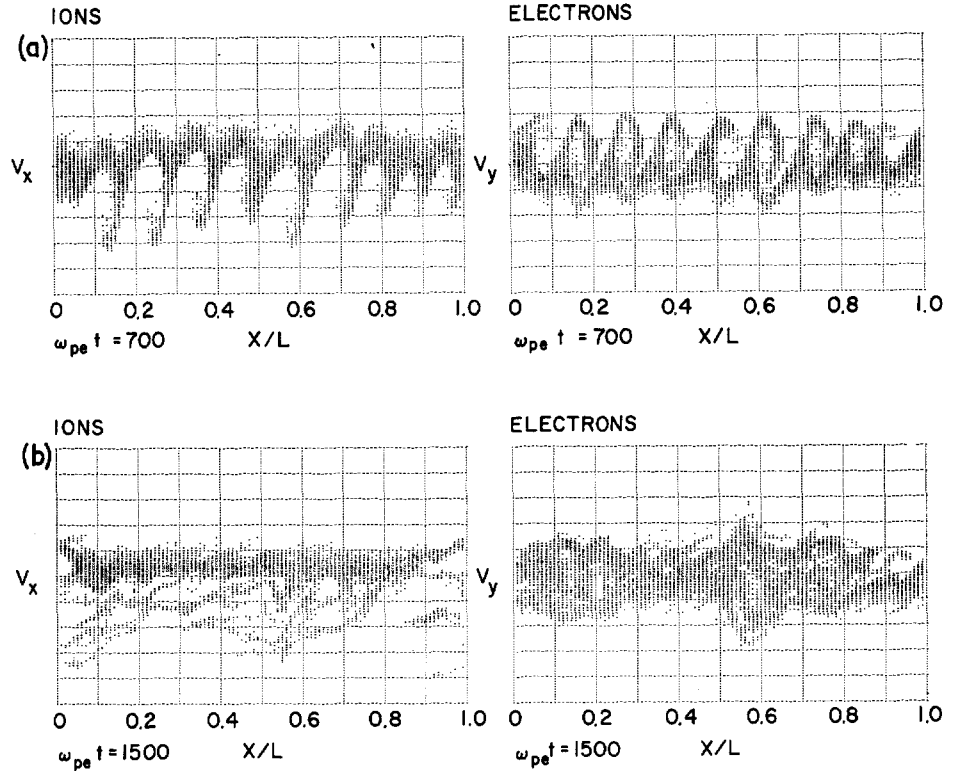


FIG. 12. Plots of electron and ion phase space at the end of the exponential stage, $\omega_{pe} t = 700$, [Fig. 12(a)] and at the end of the run, $\omega_{pe} t = 1500$, [Fig. 12(b)] corresponding to the one-dimensional simulation of Fig. 11.

The nonlinear development of the instability for the other four values of θ reported in Fig. 6 is similar to the behavior at θ_0 . In particular, the final ion temperature across B_0 is approximately the same as the final electron temperature along B_0 in agreement with trapping theory since $\phi_i \approx \phi_e$, and the fields do not damp appreciably after stabilization. Here, ϕ_i (ϕ_e) is the potential necessary to begin trapping the ions (electrons) as defined in Sec. IIIB. Whether the ions or electrons bounce first and stabilize the instability depends on whether $\tau_{Bi}/\tau_{Be} = \theta/\theta_0 < 1$ or $\tau_{Bi}/\tau_{Be} > 1$, respectively, and this is confirmed by our phase space plots. For $\theta < \theta_0$, however, a larger fraction of the free ion drift energy is transferred to the plasma than at θ_0 since $(U - \omega/k)$ is greater, and consequently, the saturation trapping width for the ions is larger. Conversely, for $\theta > \theta_0$ a smaller fraction of the free energy is transferred to the plasma. In particular for $\theta = \frac{1}{2}\theta_0$, the change in the ion streaming energy relative to its initial value is -0.89 while at $\theta = \theta_0$ it is -0.56 and at $\theta = 2\theta_0$ it is -0.29 . In all cases a very small fraction of the energy is partitioned to the electrostatic fields.

We also ran a simulation at θ_0 with the same parameters except that $T_e = 5T_i$ initially. The results were much the same as those for $\theta = \theta_0$ with $T_i = T_e$ initially, and are shown in Fig. 9. Once again the final ion temperature is approximately the same as the final electron temperature, and the fields do not damp appreciably after saturation. In this run the

change in ion streaming energy relative to its initial value was -0.68 , with almost all the energy again being transferred to the particles.

Another one-dimensional simulation was performed having $m_i/m_e = 2048$, $\omega_{pe}/\Omega_e = 1$, $T_e/T_i = 1$, $U/v_i = 10$ at $\theta = 2.25\theta_0$, with the same computer system parameters as for the simulations described above. This value of θ was chosen in conjunction with Fig. 1 to study the effect of the fastest growing modes of the instability on the simulated plasma since one might reasonably expect their effects to dominate in real plasmas. Figure 10 shows the theoretical γ vs k curve based on Eq. (4) at this value of θ . The wavenumber k is given both in terms of ρ_e and the discrete number of wavelengths N_x in the computer simulation. The dots are the observed linear growth rates of the strongly unstable modes of the system. Error bars are shown to indicate the uncertainty in measuring the individual simulation mode growth rates. The modes with lower growth rates could not be measured with sufficient accuracy to warrant their inclusion in Fig. 10. The development of these modes was dominated by the fastest growing modes. Note, for example, the interesting result that mode 17, which is linearly stable, theoretically, has been excited with a growth rate approximately equal to the sum of the linear growth rates of modes 8 and 9 (cf., Fig. 10), indicating that strong mode coupling occurs from unstable to stable regions of k space before nonlinear saturation of the dominant modes. The observed linear growth rates

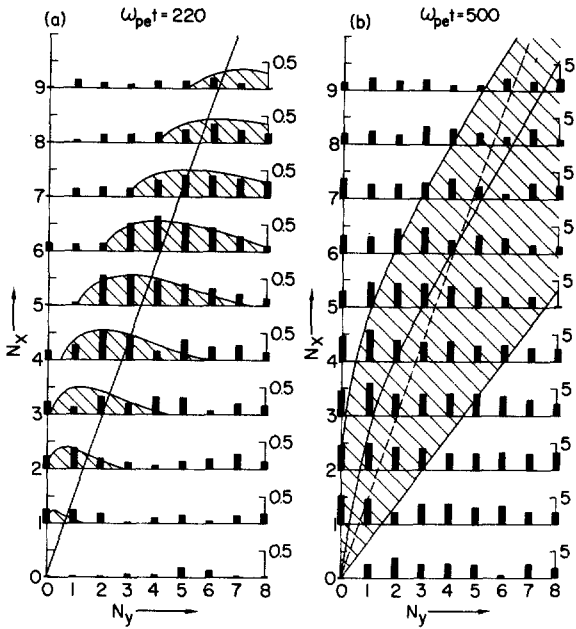


FIG. 13. The unstable linear spectrum and logarithm of mode amplitudes (solid bars) from a two-dimensional simulation of the modified two-stream instability. (a) The linear dispersion relation for γ/ω_{pi} (scale 0.5) is shown superimposed on the mode amplitudes at $\omega_{pe}t=220$ just before saturation of linear growth. (b) The shaded region of linear instability is shown superimposed on the mode amplitudes at $\omega_{pe}t=500$ following saturation of the growth due to trapping.

are once again in excellent agreement with theory. Figure 11 contains the time history of the total electrostatic energy in the unstable wave fields, the parallel electron temperature, and the perpendicular ion temperature. The electrons gain energy faster than the ions during the period of exponential development but the ion and electron temperatures are nearly the same at the end of the run (indeed $T_{\perp i}$ is a little larger than $T_{\parallel e}$). Figure 12(a) shows electron and ion phase space at the end of the exponential phase ($t=700\omega_{pe}^{-1}$). The electrons are fully trapped at this time and are responsible for stabilization while the ions are beginning to trap. In Fig. 12(b) both the electrons and ions are fully trapped and thermalized at the end of the run ($t=1500\omega_{pe}^{-1}$) since the ions have "bounced" in the potential troughs of the waves by this time. This agrees with trapping theory since $\phi_i \approx \phi_e$, the wave energy does not decay significantly after saturation, and for this value of $\theta > \theta_0$, $\tau_{B_i} > \tau_{B_e}$. After saturation the initially well-defined spectrum of waves has disappeared and the electrostatic energy at the end of the run is distributed more evenly over a broader region of k space. In this simulation the change in ion streaming energy relative to its initial value was -0.23 with less than 1% of this energy going into the electrostatic fields. This is somewhat less total energy transfer than that expected at $\theta = \theta_0$ due to the reduced saturation trapping widths for ions ($U - \omega/k$), for $\theta > \theta_0$.

We now describe the results of physically more complete two-dimensional simulations in which many waves with different projections of $\mathbf{k}$ on $\mathbf{B}$ were allowed to develop simultaneously. In our preliminary letter,¹¹ we presented the results of an initial two-dimensional run, which suffered from the coarseness of the $\mathbf{k}$ -space grid. Only a very few unstable modes were allowed due to the quantization imposed by periodic boundary conditions in a system of finite size. We have since been able to alleviate this problem and allow for a sufficiently dense spectrum in k space by appropriately reorienting the magnetic field in the calculation. In this instability the equal-effective mass angle θ_0 is quite small, being only about 1 deg for hydrogen. Thus, realistic calculations with at least several unstable modes at angles less than θ_0 are difficult to achieve when the minimum nonzero wavenumbers, $\delta k_x = 2\pi/L_x$ and $\delta k_y = 2\pi/L_y$, are comparable and the magnetic field is chosen to lie in the plane of the calculation, $\mathbf{B}_0 = (B_x, B_y, 0)$. Therefore, we established a magnetic field predominantly perpendicular to the plane of the calculation, $\mathbf{B}_0 = (0, B_y = \frac{1}{3}B_x, B_z)$. The initial ion drift was chosen in the x direction, $\mathbf{U} = (U_x, 0, 0)$ and L_x was chosen to be 64 cells ($\frac{1}{2}$ of L_y). In this geometry k_x is parallel to the drift and perpendicular to $\mathbf{B}_0$. Only k_y has a component along $\mathbf{B}_0$ and $k_{\parallel} = \mathbf{k} \cdot \mathbf{B}_0/B_0 = k_y B_y/B_0 \approx k_y/3$. Thus, we effectively tripled the number of modes in the unstable region of $\mathbf{k}$ space. The other parameters chosen for the simulation were the same as for the set of one-dimensional runs described at the beginning of this section; viz., $\Omega_e/\omega_{pe} = 3$, $T_i/T_e = 1.5$, $m_i/m_e = 512$, $U/v_e = 0.5$. In this two-dimensional run we used 2^{15} each of ions and electrons, in a system 128×64 cells with $\lambda_D = \frac{1}{2}$ cell.

Figure 13 shows the unstable spectrum at two different times during the development of the instability, at $\omega_{pe}t=220$, shortly before nonlinear saturation (see Fig. 14), and at $\omega_{pe}t=500$, well into the stabilized

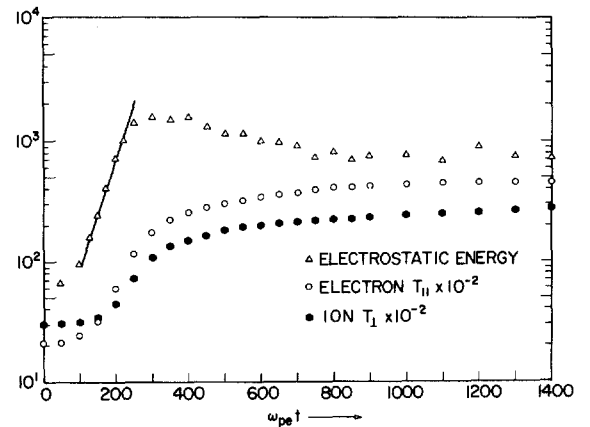


FIG. 14. Time evolution of the electrostatic field energy, parallel electron temperature and perpendicular ion temperature for the two-dimensional simulation corresponding to Fig. 13.

TABLE II. Same as the notation for Table I, except that all changes Δ refer to the values of the indicated quantities at $\omega_{pe}t = 1400$ minus the initial value.

Species	$\Delta(X \text{ Therm } En)$ $\frac{1}{2}m_e U_0^2$	$\Delta(Y \text{ Therm } En)$ $\frac{1}{2}m_e U_0^2$	$\Delta(Z \text{ Therm } En)$ $\frac{1}{2}m_e U_0^2$	$\Delta(\text{Drift } En)$ $\frac{1}{2}m_e U_0^2$	$\Delta(\text{Field } En)$ $\frac{1}{2}m_e U_0^2$	Final thermal En Final ion drift En
Electrons	0.00	0.02	0.17	0.00	...	0.31
Ions	0.10	0.02	0.00	-0.31	...	0.21
Fields	...	...	...	...	0.00	...

phase. The mode numbers are plotted on the axes of the graph for the most active region of k space. Here, $N_x=8$ corresponds to a wavelength $\lambda=16\lambda_D$ (λ_D is the initial electron Debye length) and $N_y=8$ corresponds to a wavelength $\lambda=32\lambda_D$. The straight diagonal line in Fig. 13(a) [dashed diagonal line in Fig. 13(b)] lies at $\theta=2.4\theta_0$. This angle roughly bisects the region of linear instability, shaded in Fig. 13(b), and contains the most unstable mode of the entire two-dimensional spectrum ($N_x \approx 6$, $N_y \approx 4$). At each integral value of N_x , Fig. 13(a) contains the linear growth rates scaled to the ion-plasma frequency. The scale is shown, 0 to 0.5, at the right of Fig. 13(a) and the area under the spectrum has been shaded to emphasize the theoretically unstable regime.

On the same figure, the natural logarithms of the actual individual mode amplitudes from the simulation are plotted as vertical bars. The scale, 0 to 5, is shown at the right of Fig. 13(b) so that the relative amplitudes of any two modes can be compared quantitatively. Since the natural logarithm of the mode energy is roughly proportional to growth rate in the exponential phase (all four combinations of $\pm N_x$, $\pm N_y$ have been combined in the figure), the correlation between large amplitude and large growth rate shown in Fig. 13(a) is theoretically expected. The small but nonzero excitation of modes in the linearly stable region of Fig. 13(a) is attributable to nonlinear coupling effects and to initial thermal fluctuations which are nonnegligible in two-dimensional computer simulations. Nevertheless, roughly 50 strongly unstable modes (5×5 in two dimensions times 2 from $\pm N_y$ symmetry) are seen to be in excellent agreement with the predicted spectrum in the linear regime. This is enough to insure proper representation of stochastic and turbulent phenomena and compares favorably in each direction with many one-dimensional simulation spectral resolutions.

After saturation, Fig. 13(b), the modal energy strongly couples nonlinearly to modes outside the unstable region and predominantly cascades to long wavelength as the most unstable waves break. The symmetry between $\pm N_y$ in a given value of N_x insures that no large electron drift develops either parallel or antiparallel to the magnetic field as observed for the one-dimensional simulations (cf., Table I).

Figure 14 shows the time evolution of the electrostatic energy, parallel electron temperature and perpendicular ion temperature for this simulation. As in the one-dimensional runs, the field energy does not diminish significantly after saturation and the final ion temperature is comparable to the final electron temperature (somewhat less than a factor of two smaller). The behavior of the system is similar to that reported in Fig. 2 of Ref. 11. Table II summarizes the disposition of the free energy in the initial ion-streaming motion. We note that over 30% of this energy has been converted into plasma thermal motion with the ions and electrons sharing comparably (along B_0 for the electrons and mainly perpendicular to B_0 for the ions) and a negligibly small fraction (less than 1%) has been partitioned to the electrostatic fields. A net loss of initial ion streaming energy of somewhat less than 50% might be expected based on trapping considerations since the fastest growing modes for the parameters of this simulation have $\theta > \theta_0$. At these values of θ the saturation trapping widths for ions ($U - \omega/k$), are less than $\frac{1}{2}U_0$.

The results of our one- and two-dimensional computer simulations are in good agreement with the theory of the modified two-stream instability. We have shown that this instability can be a very important plasma heating mechanism; in particular, the instability can produce large ion heating and can operate in regions of parameter space where other electrostatic instabilities are absent.

Finally, for plasma parameters such that electrostatic microinstabilities may coexist with the fluid-like modified two-stream instability (e.g., $\Omega_e < \omega_{pe}$, $T_e > T_i$, where beam cyclotron and ion acoustic modes are possible¹⁻³), we expect the final state to be dominated by the harder fluid instability. This is because the longer-wavelength fluid instabilities generally stabilize at turbulence levels above those at which the microinstabilities saturate, and the saturated levels of turbulence of the microinstabilities are not usually sufficient to alter the properties of the nonresonant fluid modes. Indeed in Ref. 2 a simulation was performed in which both the rapidly growing, short-wavelength beam cyclotron instability and a slower growing, long-wavelength fluid-like mode (the ion-ion instability in this case) were present. The short wavelength

instability grew rapidly and saturated; after saturation of the short-wavelength instability, the long-wavelength instability continued to grow unaffected and saturated at a considerably higher level than the short-wavelength mode. Thus, the ultimate level of turbulent fluctuations, and the amount of plasma heating were determined by the harder fluid-like mode, even though in this case it had a lower growth rate. As a somewhat more quantitative prediction of when the development of the modified two-stream instability dominates the final state of the plasma and is independent of faster competing microinstabilities, we note that the fast exponential phase of both ion acoustic modes and beam cyclotron modes (which have been shown to behave basically like ion acoustic modes² for $\Omega_e \ll \omega_{pe}$) ends with the onset of ion trapping. Thus, since the phase velocity of these modes is $\sim c_s$ (ion sound speed), while the phase speed of the nonresonant fluid modes is $\sim U$, the fluid instability dominates the final state if $U^2/c_s^2 \gg 1$ initially.

V. APPLICATIONS

In this section we review some experiments in which there is relative drifting between electrons and ions across a magnetic field such that the modified two-stream instability may be present. It is interesting to speculate on the possibility of actually inducing such drifts for the specific purpose of plasma heating by the modified two stream instability.

A. The Experiment of Ashby and Paton

In the experiment of Ashby and Paton⁷ a low-beta hydrogen plasma is injected into a solenoidal magnetic field. The ion gyroradius is of the same order as the radius of the plasma column so that in the outer portion of the plasma the gyromotion, in effect, causes the ions to act like a beam directed across a magnetic field (cf. Fig. 1 in Ref. 7). The authors observed turbulent fluctuations and attributed their results to the modified two-stream instability.

B. $\mathbf{E} \times \mathbf{B}$ Plasma Rotation Experiments

If a radial electric field exists in a plasma, the plasma will tend to rotate due to the $\mathbf{E} \times \mathbf{B}$ drift of electrons and ions. The ions, due to their much larger mass, then feel an outward centrifugal force which is much larger than that experienced by the electrons. This difference in centrifugal forces leads to a relative drift between electrons and ions in the azimuthal direction equal to (cF_e/eB_0) where F_e is the difference between the ion centrifugal force and the centrifugal force on the electrons, $F_e \cong m_i c^2 (E_r/B_0)^2/r$. This relative drift can then lead to instability. We are aware of two experiments which utilize this effect.^{18,19} The first experiment performed was that of Ioffe *et al.*,¹⁸ in which the radial electric field was caused by the space charge resulting from an ion beam which was

injected axially down the center plasma column with B_0 also axial (actually a mirror). These authors observe that this method leads to a plasma with hot ions. Mikhailovskii and Tsypin⁵ (see also Krall and Liewer⁴) have pointed out two instabilities which appear to be capable of explaining this result; one is the modified two-stream instability; the other is a drift mode resulting from the radial density gradient in the plasma. The experiment of Alexeff *et al.*^{19,20} is similar to that in Ref. 18 but utilizes an electron beam, rather than an ion beam, to produce the radial electric field. These authors also note a number of other possible instabilities in addition to the two noted by Mikhailovskii and Tsypin. It appears that in this experiment the drift-mode instability is the dominant mechanism. Which instability dominates in a particular experiment of this type depends on the precise experimental conditions (e.g., strength of the radial density gradients, etc.).

C. Anomalous Absorption of Magnetosonic Waves

In the turbulent heating experiment of Babykin *et al.*²¹ a plasma is located in a magnetic mirror machine. Surrounding the contained plasma is a rf tank coil which sets up an alternating magnetic field ($f \sim 9$ MHz) $B_\perp$ which is superimposed on the confining mirror field. The alternating magnetic field produced by the rf circuit excites a radially propagating magnetosonic wave. The magnetic field gradient in the wave is related to the electron current flow in the wave by Ampere's law, $-4\pi N_e U = c\partial B_\perp/\partial r$. If this current is sufficiently strong, it can excite instabilities in the plasma and result in the observed heating of the plasma. Of the possible instabilities excited, the modified two-stream instability appears to be the most likely explanation of their results. A similar method can be used to heat tokamaks and other fusion devices. In this connection it should be pointed out that the modified two-stream instability is ineffective unless $U \gtrsim V_A (1+\beta_e)^{1/2}$ [cf. Eq. (9)]. Since $U > v_i$ is also required for instability, an upper limit on the ion beta is implied, $(1+\beta_e)^{1/2} V_A > v_i$ or $\beta_i < 2(1+\beta_e)$, which is not much of a restriction for proposed low-beta fusion devices. Furthermore, from Ampere's law we obtain an estimate for the relative electron ion cross-field drift in the form

$$U \sim (m_i/m_e)^{1/2} V_A (B_\perp/B_0) (k_0 c/\omega_{pe}), \quad (31)$$

where k_0 is the wavenumber of the magnetosonic wave. Combining Eq. (31) with Eq. (9) we obtain

$$(m_i/m_e)^{1/2} (B_\perp/B_0) \gtrsim (\omega_{pe}/k_0 c) (1+\beta_e)^{1/2} \quad (32)$$

which puts a restriction on the amplitude and frequency of the driving rf magnetic field.

Another interesting experiment is that of Grigor'eva *et al.*²² who excite a magnetosonic wave and observe wave absorption accompanied by plasma heating.

Such anomalous absorption of magnetosonic waves may also be an important process for heating ions in the solar wind.²³

D. Cross-Field Ion Beam

Probably the simplest way in which relative cross-field electron-ion streaming can occur is by shooting an energetic ion beam across a magnetic field. If the gyroradius of the beam is much larger than the size of the magnetic field region, the beam will not be influenced by the magnetic field and will move straight through. This type of situation has been realized in the recent experiments of Barrett *et al.*¹² whose results are explainable by the development of the modified two-stream instability.

E. Collisionless Shock Waves

The electron drift in a collisionless shock may be estimated from the equation $c\nabla \times \mathbf{B} = 4\pi\mathbf{J}$ to be $U \sim (m_i/m_e)^{1/2} V_A (c/\omega_{pe} L_s)$, where L_s is the shock thickness. For perpendicular shocks L_s is typically $\sim 10c/\omega_{pe}$ so that in a hydrogen plasma $U \sim 4V_A$ and the stabilizing electromagnetic effects, Eq. (9), become strong. For oblique shocks L_s is longer $\sim c/\omega_{pi}$. Thus, conditions for excitation of the modified two-stream instability in oblique shocks are more favorable, and this instability is a possible mechanism for providing anomalous resistivity in oblique shocks.

F. Anomalous Momentum Coupling between Interpenetrating Plasmas

This instability may also be an important mechanism for collisionless momentum coupling between interpenetrating plasma streams. A number of experiments relevant to this topic have been performed.²⁴⁻²⁶ In these experiments a blob of plasma is created and expands outward through a background plasma (e.g., in Refs. 25 and 26 the expanding plasma is created by laser irradiation of a solid pellet). A question of prime importance to the dynamics of such a situation is the following: Will the expanding plasma simply free stream through the background without affecting it, or will the expanding plasma couple to the background through some anomalous friction mechanism thus transferring momentum from the expanding plasma to the background? Coupling can be achieved via instabilities which feed on the free energy inherent in the streaming of one plasma through another. Figure 15 schematically shows the distribution function for electrons and ions for two interpenetrating plasmas drawn in the rest frame of the electrons. It is evident that two types of streaming instabilities are possible: ion-ion instabilities and electron-ion instabilities. Since the momentum is carried mainly by the ions due to their greater mass, ion-ion instabilities provide a direct mechanism for momentum coupling. Electron-ion instabilities, such as the modified two

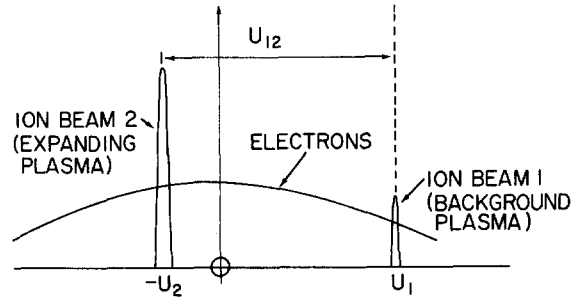


FIG. 15. Schematic of the distribution function for the electrons and ions of two interpenetrating plasmas drawn in the rest frame of the electrons.

stream instability, can also couple the momentum from one ion beam to another but in a less direct fashion: Any friction force between one ion component and electrons must manifest itself as a force on the other ion component caused by the macroscopic electric field which develops in order to maintain over-all macroscopic quasineutrality of the flow ($n_e \approx n_{i1} + n_{i2}$). It is clearly of interest to compare the condition for the magnetized ion-ion instability²⁷ with that of the modified two-stream instability. In the laboratory experiments cited²⁴⁻²⁶ the density of the expanding plasma (n_{i2}) appreciably exceeds the density of the background (n_{i1}). In this case, the marginal instability condition for the magnetized ion-ion instability is $U_{12} < B_0(1 + \beta_e)^{1/2} / (4\pi n_{i2} m_{i2})^{1/2}$. On the other hand, the relevant velocity for a modified two-stream instability between ion beam 2 and the electrons is U_2 . Since U_2 is less than U_{12} by about (n_{i1}/n_{i2}) , conditions for the onset of the modified two-stream instability are less stringent than those for the ion-ion instability. This is an important consideration in some situations where $n_{i2} \gg n_{i1}$, and the condition for onset of the direct ion-ion instability is not satisfied.²⁸

VI. SUMMARY

In this paper we have presented a detailed analytical and numerical study of the fluid-like, modified two-stream instability, which has a characteristic frequency and growth rate comparable to the lower-hybrid frequency. The linear theory was discussed in both the electrostatic and fully electromagnetic cases (Sec. II), and we presented a detailed numerical study of the dependence of the linearly unstable roots of the dispersion relation for a wide range of plasma parameters. Our nonlinear theory (Sec. III and the appendix) includes discussions of (1) quasilinear theory, (2) trapping, (3) a fully nonlinear scaling law which shows how results scale with m_i/m_e , and (4) the effect of cross-field vortex like motion caused by turbulence induced $\mathbf{E} \times \mathbf{B}$ drifts (appendix). One- and two-dimensional computer simulations with dense $\mathbf{k}$ -space spectra were presented and related to theory in Sec. IV.

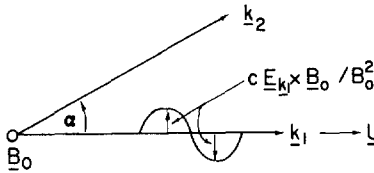


FIG. 16. Schematic of the situation described in the Appendix concerning the nonlinear effect of electron $\mathbf{E} \times \mathbf{B}$ motion.

The results of the study give strong indication that this instability can be a very important turbulent heating mechanism in plasmas that have a relative drift of electrons and ions perpendicular to a magnetic field such that

$$1 < U/v_i \lesssim [2(1+\beta_e)/\beta_i]^{1/2}.$$

Such a situation is realized in a wide variety of practical applications as discussed in Sec. V. In particular, the instability efficiently heats the ions of a plasma (mainly perpendicular to $\mathbf{B}_0$) to temperatures comparable to those to which it heats the electrons (along $\mathbf{B}_0$), in times of the order of ω_{LH}^{-1} and results in a final state characterized by $(T_{ix}/m_i)^{1/2} \approx (T_{ex}/m_e)^{1/2} \sim \frac{1}{2}U$, where U is the initial drift speed. In this regard we have suggested that the instability may be taken advantage of to heat ions in low β_i fusion devices, e.g., tokamaks and magnetic mirror machines, by pulsating the magnetic field at a low frequency so as to induce cross-field drifts with $U > v_i$.

APPENDIX: NONLINEAR EFFECT OF ELECTRON $\mathbf{E} \times \mathbf{B}$ MOTION

Consider a wave $\mathbf{k}_1$ as shown in the plane perpendicular to $\mathbf{B}_0$ in Fig. 16. The component of the field $\mathbf{E}_{\mathbf{k}_1}$ which is perpendicular to $\mathbf{B}_0$ leads to $\mathbf{E} \times \mathbf{B}$ drifts of the electrons also shown in Fig. 16. Since these drifts are perpendicular to $\mathbf{k}_1$, they do not affect the electron density perturbation at $\mathbf{k}_1$ (except insofar as they are responsible for the polarization drift). Now consider another wave $\mathbf{k}_2$ (cf. Fig. 16). It is clear that since $(\mathbf{k}_1)_\perp$ and $(\mathbf{k}_2)_\perp$ are not parallel, that the $\mathbf{E} \times \mathbf{B}$ drifts due to $\mathbf{k}_1$ will tend to shear apart and destroy any electron density perturbation produced by $\mathbf{k}_2$ (and vice versa). The time necessary for the wave at $\mathbf{k}_1$ to shear apart the wave at $\mathbf{k}_2$ is

$$\tau^{-1} \sim c(E_{\mathbf{k}_1}/B_0)k_2 \sin \alpha, \quad (\text{A1})$$

where α is the angle between $\mathbf{k}_1$ and $\mathbf{k}_2$. We can estimate the amplitude of $\mathbf{E}_{\mathbf{k}_1}$ at which this effect becomes important by setting $\gamma\tau \sim 1$, where γ is the growth rate of wave $\mathbf{k}_2$. Taking $\gamma \sim \omega_{pi}(1+\omega_{pe}^2/\Omega_e^2)^{-1/2}$ and $k \sim \omega_{pi}(1+\omega_{pe}^2/\Omega_e^2)^{-1/2}/U$, we obtain the amplitude of $\mathbf{E}_{\mathbf{k}_1}$ necessary to affect the wave at $\mathbf{k}_2$

$$(\sin \alpha) E_{\mathbf{k}_1} \sim UB_0/c. \quad (\text{A2})$$

Actually, there are not just two waves but a whole spectrum of waves so that Eq. (A2) should be replaced by

$$\langle \sin^2 \alpha \rangle \langle E^2 \rangle \sim (UB_0/c)^2, \quad (\text{A3})$$

where $\langle \sin^2 \alpha \rangle$ is a measure of the angular width of the wave spectrum in the plane perpendicular to $\mathbf{B}_0$. This effect will be important if the above value, Eq. (A3), is less than that at which the instability stabilizes by trapping, $(eE/km_i)^{1/2} \sim U$. Putting this in Eq. (A3), we obtain

$$\langle \sin^2 \alpha \rangle \lesssim (\Omega_i/kU)^2. \quad (\text{A4})$$

Since the term on the right-hand side of (A4) must be small (we have taken the ions to be unmagnetized so that this assumption is implicit), this $\mathbf{E} \times \mathbf{B}$ effect is probably the first nonlinear effect that comes into play. We conjecture the following sequence of events: The waves grow linearly until (A3) is satisfied. At this point, the faster growing modes, which are at lower α angles start shearing apart the slower growing modes at larger α . Thus, $\langle \sin^2 \alpha \rangle$ decreases while $\langle E^2 \rangle$ continues to increase until growth is finally arrested by trapping as discussed in Sec. IIIB. At saturation $\langle \sin^2 \alpha \rangle \sim (\Omega_i/kU)^2$. Thus, the main effect of this $\mathbf{E} \times \mathbf{B}$ nonlinearity may be to cause a collapse of the spectrum to the $\mathbf{U}-\mathbf{B}_0$ plane.

The effect just discussed may be described in a more formal fashion by utilizing the test wave theory for plasma turbulence developed in Refs. 29 and 30.

* On leave of absence from Cornell University.

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Spectroscopic Observations of Turbulence in a Fast Theta Pinch

W. D. DAVIS

Department of Physics and Astronomy, University of Maryland, College Park, Maryland 20742

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The spectrum of fluctuating fields in a collisionless magnetized plasma produced in a high voltage theta pinch is studied using satellite line techniques. Differences in the turbulent spectrum exist depending on whether the initial magnetic field is parallel or antiparallel to the driving magnetic field with the possible identification of electron cyclotron modes in the antiparallel case and electron plasma oscillations in the parallel case. By including quadratic Stark shifts, difficulties in interpretation of an earlier satellite experiment are explained. Predictions of various instability theories are tested experimentally with the aid of additional diagnostic methods.

I. INTRODUCTION

The behavior of the magnetic implosion in the Maryland turbulent heating experiment changes markedly depending on the polarity of the bias field relative to the imploding magnetic field.¹ When the bias field is parallel to the imploding field (parallel case), a diffusive penetration occurs with very little plasma motion until the magnetic disturbance is nearly halfway into the tube. Some electron heating and a significant amount of ion heating² occur during the implosion. For the bias field antiparallel to the imploding field (antiparallel case), a substantial amount of plasma is quickly "picked up" along with the rapid formation of a shock. Strong electron and ion heating are observed in this case.

In view of the large differences in macroscopic properties between the two cases, it was felt that observations of differences in the turbulent spectrum of the plasma might also be possible, and if so, would lead to a better understanding of the macroscopic behavior. This possibility was investigated using helium plasma satellite lines of Baranger and Mozer.³

Briefly, this technique depends on the stimulation of a forbidden atomic transition by the oscillating electric

fields of plasma waves. The resulting second-order line spectra consists of two satellite lines, one appearing on either side of the forbidden line and separated from it by the frequency of the plasma wave Ω . For isotropic turbulence, the ratio of the intensity of the satellite lines to the allowed line is to second order

$$S_{\pm}^{BM} = \hbar^2 R_{ll'} \langle E^2 \rangle / 6m^2 e^2 (\Delta \pm \Omega)^2, \quad (1)$$

where Δ is the spacing between the allowed and forbidden lines, E is the electric field of the plasmon, and $R_{ll'}$ is the radial matrix element for the plasmon transition. Therefore, the frequency spectrum and average electric field strength of the plasmons can be obtained from the line profiles of the satellite and allowed lines.

II. TURBULENT HEATING EXPERIMENT

The Maryland turbulent heating experiment consists of a 46-cm diam Blumlein-fed theta pinch with a 100-cm coil length, described in Ref. 1. The initial plasma parameters are: $T_e \approx T_i \approx 1$ eV, $N_e \approx 10^{18}$ cm⁻³, $\beta \ll 1$ where T_e , T_i are the electron and ion temperatures, N_e is the electron density, and $\beta = 8\pi N_e kT / B^2$, where B is the initial magnetic field. A 6-m Torr filling of helium was used in this work.